
Machine Learning System for Fresh Food Demand Forecasting in Airline Catering

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What I will cover

01 Problem

02 EDA

03 Models

04 Results

Inaccurate existing forecasting model

52%

forecast accuracy of the current
method

48%

of loaded fresh food is never sold

24 h

shelf life — no second chance to sell

Why the current method cannot be improved



Two layers, one number – and no way to tell which layer made the error.

Four objectives

I

Explore the data

2

Develop a model

3

Adapt to
production

4

Enable business rules

Four sources, one mart

I SOURCES

Order summary

Passenger counts

Flight schedule

Sales

II TRANSFORM · DBT

Staging

Intermediate

Marts

III RESULT

`mart_fresh_`
`food_order_sale`

the single input to the forecasting pipeline

The dataset at a glance

52,830

observations after cross-validated feature construction

86

flight numbers, across roughly 58 origin-destination pairs

10

fresh food SKUs across 6 categories

4

months: November 2025 – February 2026

Most flight–product pairs sell nothing



Share of all flight–product observations

6.99

skewness — extreme right tail

104.6

kurtosis — very sharp peak at zero

0–42

units, 33 distinct target values

Task definition and how success is measured

PREDICTION TASK

Sales quantity per
flight × product

TECHNICAL METRIC

MAE

in units, robust to outliers

BUSINESS METRICS

Accurate predicted = actual

Relative Accuracy $\frac{\min(\text{predicted}, \text{actual})}{\max(\text{predicted}, \text{actual})}$

Potential waste predicted > actual

Potential lost sale predicted < actual

Five features, one engineered hierarchy

I FEATURE SET

item_id product identity

route origin–destination pair

day_period night / morning / day / evening

pax_bin load category

hist_avg hierarchical historical average

II HIST_AVG · THREE-LEVEL FALLBACK

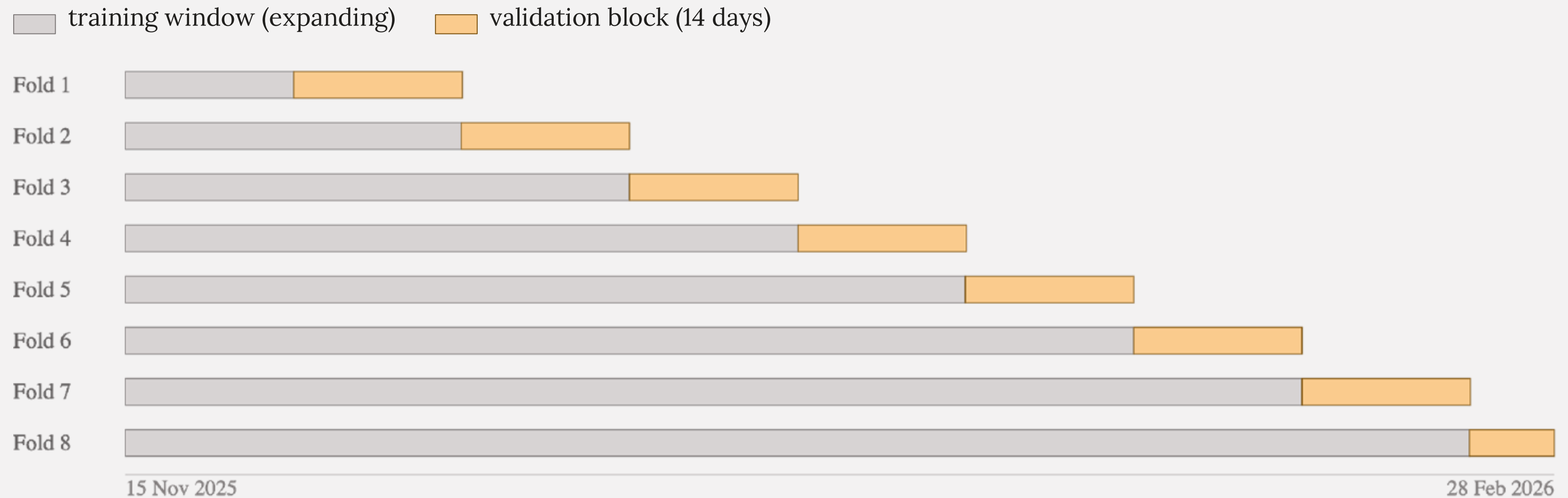
Level 1 item × route × day_period 71.65%

Level 2 item × route × is_night 87.90%

Level 3 item × day_period 100%

Coverage at each level; level 3 is the fallback for new routes.

Historical averages must never see the future



Two reference models before CatBoost

MODEL 1

Historical mean baseline

round(hist_avg), clipped at zero. No training.

0.482	62.5%
MAE	exact accuracy
21.8%	15.7%
waste	lost sales
66.8%	
relative accuracy	

MODEL 2 · DROPPED

K-nearest neighbours

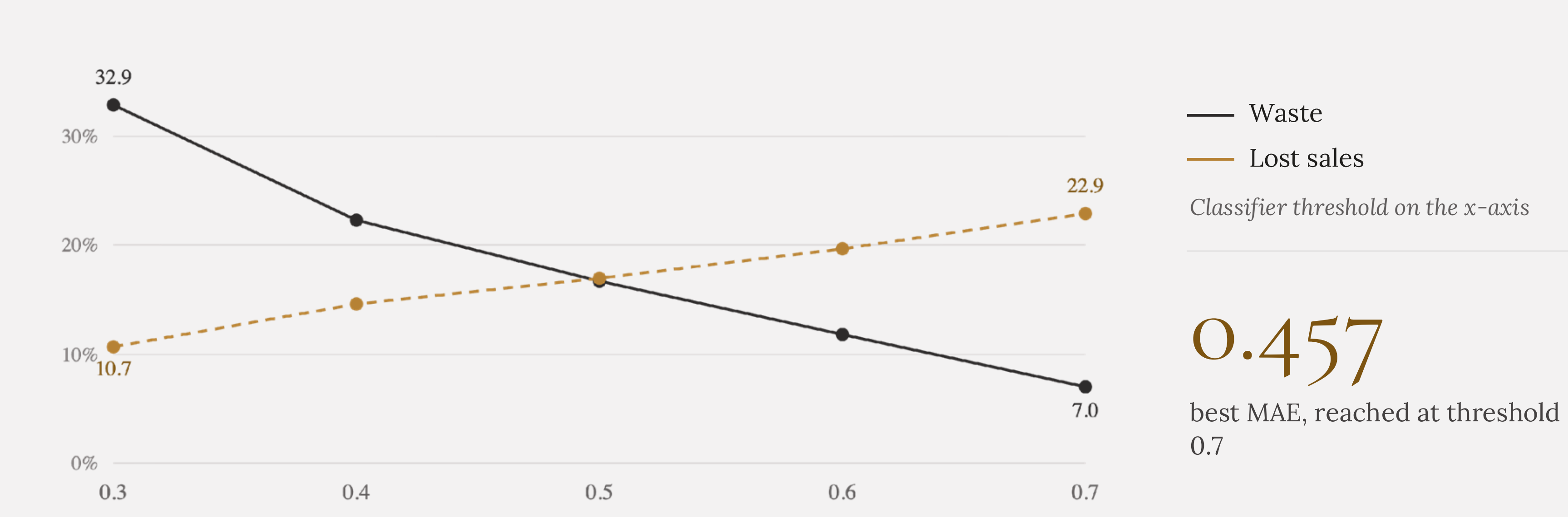
Average the most similar individual records, not one group average.

0.469	64.3%
MAE (k = 7, t = 0.7)	exact accuracy
16.3%	19.4%
waste	lost sales
68.0%	
relative accuracy	

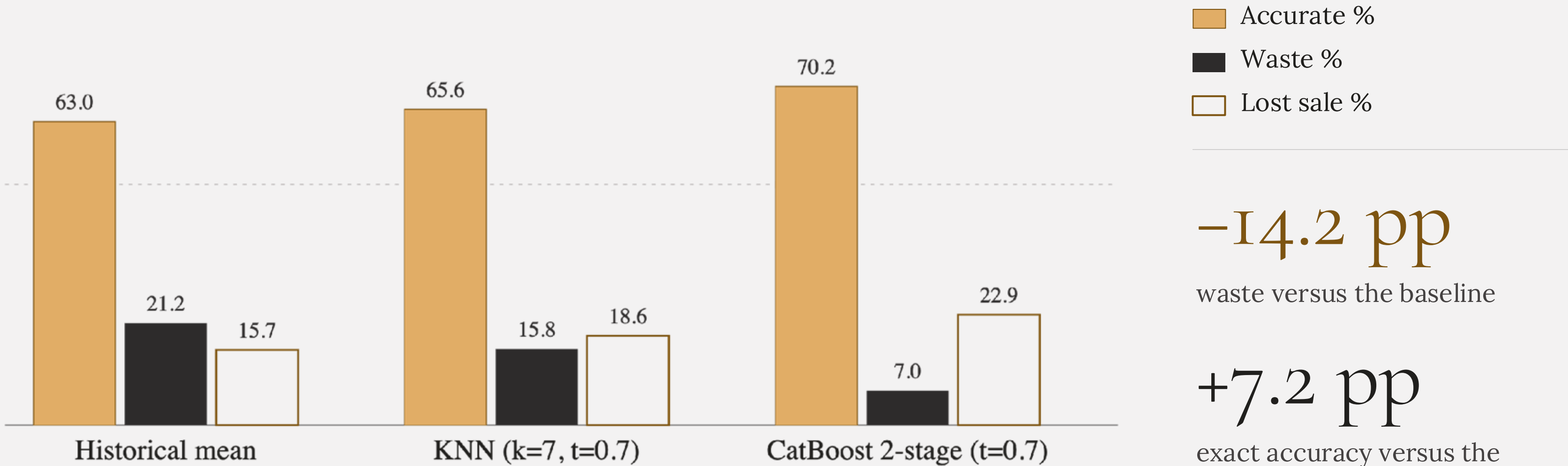
Two-stage CatBoost: decide, then quantify

I INPUT	II STAGE 1 · CLASSIFIER	III STAGE 2 · REGRESSOR	IV OUTPUT
5 features item, route, day_period, pax_bin, hist_avg	Will this product sell at all? outputs P(sale), F1 objective · P < threshold → predict 0	How many, given a sale? MAE loss, trained on non-zero records only	Loading recommendation
	72.8% relative accuracy	66.9% precision at t = 0.7	

Waste and lost sales move in opposite directions



Waste falls from 21.2% to 7.0%



-14.2 pp
waste versus the baseline

+7.2 pp
exact accuracy versus the baseline

The threshold is a business control, not a hyperparameter

MODE A

low_missed_sales

threshold = 0.4

22.3%	14.6%
waste	lost sales

Protects revenue and passenger experience — peak season, premium routes, new product launches.

MODE B · RECOMMENDED DEFAULT

low_wastage

threshold = 0.7

7.0%	22.9%
waste	lost sales

Prioritises cost control and sustainability targets. Best MAE and best exact accuracy on the test set.

From notebook to a production service



Human-in-the-loop deployment gate

A degradation check blocks registration; the active model keeps serving.

Everything configurable in YAML

A new operating configuration needs no code change.

Keeping the model correct after deployment

01	Scheduled retraining	fixed cycle, expanding window, audit record per run
02	Triggered retraining	new routes, catalogue or fleet changes
03	Performance monitoring	metrics stored per model version
04	Drift monitoring	pax_bin, route and day_period distributions
05	Alerting	degradation surfaces for human review

Demo

The inference service, running.

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Contributions, limitations, next steps

I CONTRIBUTIONS

A two-stage model for aviation retail data

Forecast separated from the business rule layer — error is attributable

A production system with a configurable business parameter

Waste 21.2% → 7.0% on held-out data

II LIMITATIONS

Four months — no year-over-year seasonality

One airline, one hub; pax_bin is A320/A321-specific

Lost sales rise to 22.9% at the recommended threshold

III FUTURE WORK

Per-SKU threshold calibration

Monitoring dashboards, technical and business

Prediction logging against actual sales

Extend observation to a full calendar year

Thank you

Questions are welcome.

63.5% zeros

two-stage CatBoost

MAE 0.450

accuracy 70.2%

waste 21.2% → 7.0%